COM6016: Cyber Threat Hunting and Digital

Forensics - Research Report

Done by –

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INTRODUCTION

IOT

The Internet of Things Refers to a large network of Physical devices that can be wirelessly connected to an Internet Network, to be more precise IoT consists of objects that are equipped with Sensors, Embedded with Software and Other Technologies that allow them to transmit and receive data with Other Devices and System.

IoT Forensics

IoT Forensics is a Branch of Digital Forensics that emphasizes Investigating and analyzing data From IoT devices involved in Cybercrime. It Involves Collecting, Preserving, and examining digital evidence from these devices to have a better understanding of Security Breaches or any incidents where IoT data is relevant.

Why is IoT Forensics Important?

In the modern world where the usage of IoT Devices has increased Drastically From home appliances to industrial sensors, there is a potential risk of cyberattacks and Security Breaches that are more likely to happen. This is where IoT Forensics plays a Crucial role. Large volumes of data are being stored and exchanged by IoT devices, making them easy targets for criminal activity, data breaches, and cyberattacks. IoT Device often stores data logs that can serve as vital digital shreds of evidence for a cybercrime, unauthorized access, or even physical crimes. Investigators utilize the pieces of evidence collected from various like cloud storage, network data, and device logs to examine and reconstruct data trails to bolster their cases. Not

only IoT Forensic can store data but also can analysis evidence or any digital traces left behind to figure out how the attack has occurred and identify vulnerabilities so that this can be present shortly. In this way, IoT forensics helps the industrial sector, health, and other sectors secure and can resist and bounce back from cyber threats.

Tools Used in IoT Forensics

IoT Forensics uses a variety of tools and methods to gather, examine, and store data from IoT networks and devices. These tools help forensic analysts improve cybersecurity overall, assist forensic investigators in comprehending security occurrences, and collect data for legal purposes. The following is a collection of frequently used IoT forensics tools, arranged according to their functions:

1) Autopsy: -

It is an open-source platform for Digital forensics that is used in various forensic investigations which includes IoT Forensics. It has plugins and modules which makes it adaptable for data sources as well as IoT devices. It can get data from storage media from memory cards, SSD cards, USB drives, or internal storage within IoT devices and it is suitable to recover data as well. When it comes to examining the files it can identify deleted files from the past, images, relevant logs, configuration of files, and other data. Logs are found when Autopsy analyses the files where that contain valuable information about the device activities, and network interactions which are crucial during the time of security incidences like unauthorized access or any data breaches.

2) Wireshark: -

Wireshark is a network protocol analyzer that is used in IoT forensics for capturing data and analyzing network traffic. It is a valuable asset in IoT forensics because IoT devices communicate over the network through Wi-fi and Bluetooth, and these interactions are quite essential making it a resourceful asset for understanding the behavior of IoT Devices and protect from any Security Threat. Usually, IoT devices use a variety of protocols for communication with cloud servers and other IoT devices Wireshark analyses it and decodes it as well. Some of the protocols used by IoT Devices are HTPP, MQTT, and CoAP. By looking through the network patterns, IoT forensics can identify any suspicious activities and indicate any malicious activities. This includes when is there sudden unusual traffic in the network causing a DOS attack, unexpected data flow, or any abnormal deviation in normal communication.

3) FTK Imager:

Forensic Tool Kit Imager is an imaging tool that is used to collect shreds of evidence to create Forensic images of various storage media without tampering with the original Data, it is a valuable software in IoT Forensics due to its capabilities for analyzing and preserving data and It is Especially found in IoT devices like Tablets, Smartphones, and Smartwatches. FTK Imager is often used to acquire data from Removable storage such as SD cards used in smartphones, Cameras, and Drones as well. Internal storage is when IoT Devices have internal memory that can be accessed directly by connecting a USB device.

Two Significant Challenges during IoT Forensics and Its significances

IoT devices these days have a shortage of storage capacity and only sustain for a short period so they may overwrite older data or delete it due to

constraints in storage to preserve the privacy and efficiency of the device. Additionally, IoT Devices are frequently stored in a volatile memory which can be removed easily in case of which the device is turned off or reset. Most often cloud-based storage is present in most of the modern IoT devices as it makes it more difficult to access or recover this data from any other external servers due to the potential data retention policies of the service provider. This Data Volatility can significantly impede Forensic investigators, Due to this instability, the investigation may be affected if there is a delay in the collection of data. The quality of forensic analysis also becomes difficult due to the limitation of data retention and makes it even more challenging to get the prior occurrence of the file as a result investigators need to hire an advanced data recovery technique like network traffic analysis to recreate the digital timeline of the events that occurred. If important evidence is lost, it could affect the outcome of the analysis report which potentially weakens a legal case.

Another main challenge is that IoT devices nowadays vary from brand and manufacturer to manufacturer as well because in terms of hardware which might be different, operating systems again different from different manufacturers for instance apple IoT devices their operating system, whereas Amazon Alexa and Google Home has a different Operating system and configurations, communication protocols, may store the data differently or rely on cloud services.

Because of the various device types, operating systems, and configurations, the investigators need different tools and methods to analyze and acquire the data. For example, in a criminal case, data in thermostats was critical in finding the timeline and the chronologic events of the crime. However, due to the nature of its storage of data and communication protocol, it will take some time to reverse engineer the process as this will bring delay in further investigation as well. Not only this, the time taken to use a particular tool according to the device preference and having the specific knowledge as well

to do so, increases the cost of the investigation and results in loss of crucial evidence.

Challenges are addressed in practice.

In Limited storage cases as we discussed in the above paragraph to reduce any loss of data investigators should give importance to real-time data collection as they need to act quickly to eliminate the risk of data being overwritten or erased and their main aim should be to collect the data from the memory dump when the device is still alive before turning it off. Using FTK imager tool and other relevant tools to extract and preserve data. When it comes to cloud storage which most IoT Devices use. But to access this data is a real issue for investigators as they need approval from the court through legal agreements from the cloud service provider to access data, clear logs, and backups. When the data recovery is not feasible, investigators can use network traffic analysis to gather evidence and monitor the network and the traffic investigators can use Wireshark as well. As tools are important in extracting data and recovering data from cloud data where timing plays an important role in any loss of shreds of evidence.

The development of Forensic tools has increased as the need to handle a wide range of different IoT devices and their new configurations. Cellebrite UFED a well-known in mobile forensics, has recently developed software that can support data analysis and extraction from IoT Devices like wearables smartwatches, and smart home devices. Investigators must use a multi-tools approach so they might use one tool

to extract from thermostats and another for voice assistants like Amazon Alexa or Google Home. Investigators need to gain some specific knowledge for specialized training to handle specific IoT systems when

it comes to different operating systems. Apple configurator for Apple devices and Android devices is Android Device Bridge.

Part B: Digital Forensics and Natural Language Processing.

Introduction and Paper Reference

Performance Evaluation of a Natural Language Processing Approach Applied in White Collar Crime Investigation

https://arxiv.org/abs/1609.00992 - Reference link about the chosen paper.

Digital Forensics Problem Being Addressed and Its Importances

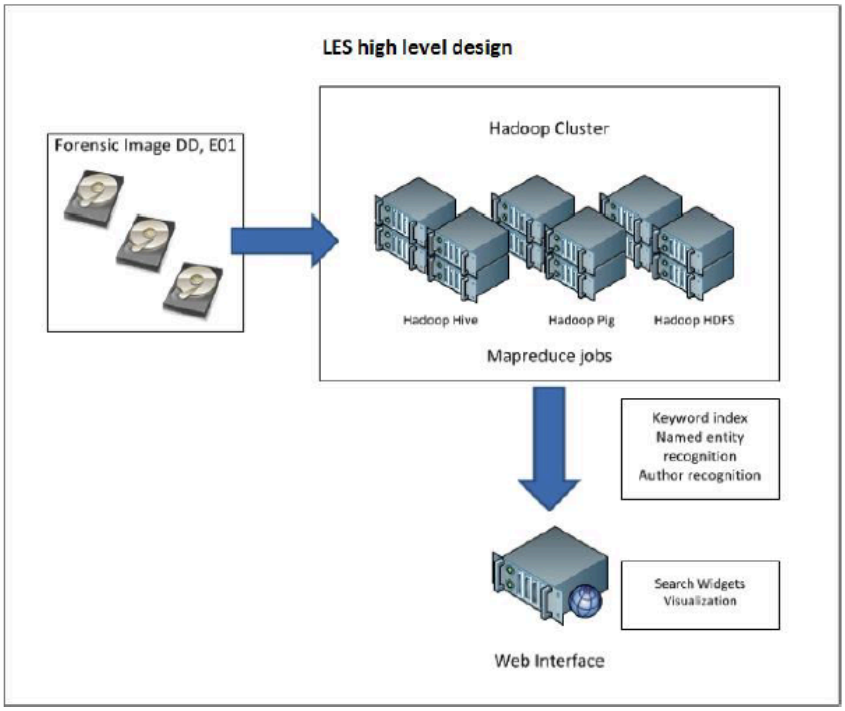
Everything since then and data-driven applications have led to increasing amounts of data consumption, as this is where new challenges have arrived. Law agencies like the Crime Investigation Department of the Customs Force in a European country have to analyze around 3.5 terabytes of data per case. This Data can include Financial Records, Emails, Official Documents, and Social media posts making it challenging to get relevant information. These forensics tools are unable to comprehend these big data due to the large volume and complexity of the data which delays the process of the investigation.

White crimes are prone to happen quite a lot these days due to vulnerabilities present in the systems and servers.CID is concentrated on the criminal prosecution of Economic, insider trading, bankruptcy fraud, property fraud, and money laundering which are general white-collar crimes that happen quite often. So recently CID has come up with a newly developed tool called LES Tool is used to exploit very large and complex data not only it also comes in handy in extracting key information such as name entities, key events, and relationships and it can also timeline of events associated with network activities and files which helps the investigators to reconstruct the sequences of events. With the rise of digital crimes, the need for efficient tools to assist

investigators in the context of white-collar crime investigations is important.

How NLP Has Been Applied to the Problem

We previously talked about the Problems and how it's impacted the Investigation. Still, now we discuss how Natural language processing Techniques are implemented to solve complex fraud cases and analyze large data pieces of evidence files.LES tool as we know it's been a game changer for the investigator but let's get to know more about its main characteristics, It uses the NLP technique to improve finding evidence which makes it even easier to investigate as well, It Runs on an Apache Hadoop platform and as we discussed that its Ability to handle large amounts of data and it could also Visualize the evidence that is found, lastly it has advanced search capabilities in simple web-based GUI. There is a software component called MapReduce, It is a built-in LES tool for text extraction and text indexing. The combination of techniques in which NER and information extraction are used to extract useful information and key entities such as names, locations, and organizations from documents as it uses a distance-based approach to relationship extraction.



To make data handling simpler LES tool investigating process is divided into several steps. In the Initial stage, Digital Evidence is collected, gathered, and often formatted in a forensic format which is raw dd files or encases. During the Extraction phase, The Hadoop Cluster mounts the data, checks the integrity, and arranges it into a binary data system. The Indexing Phase data is transformed into searchable streams which enable efficient and easy access to the data and during the ending stages which is the final Relationship Extraction and Analysis Phase the relationship between entities is identified, processed, and imported into the LES Hadoop Elastic search environment where the investigators can view and examine and analyze this Connections and dataset.

Successful application of NLP

The LES tool in the NLP application is highly efficient and time-efficient for Investigators as well. Considering the performance perspective between the forensic toolkit and the LES Tool. In processing time of the testing dataset in FTK takes about 10 hours and 54 minutes and for LES Tool the total processing 1hour and 24 minutes including text extraction as well and generating a searchable keyword index based on the testing datasets which is done in several minutes and generating NLP NER databases in 38 minutes. Overall LES tool is 6x times faster than the FTK tool while testing the same data tests. Here below image shows the response time of keyword searching FTK shows Slows response time when compared with the LES tool when using a single keyword.

Table 1. Response time of single keyword search

Document	*Keywords*	*FTK*		*LES*	
		Response Time(s)	*Retrieve Time(s)*	*Response Time(s)*	*Retrieve Time(s)*
D-195	discontinue	6	27	0.005	< 0.5
(.msg)	fraud	3	41	0.078	< 0.5
	revenue	6	287	0.119	< 0.5
D-550	scrubbing	6	17	0.081	< 0.5
(.ppt)	blacklists	5	14	0.099	< 0.5
	violation	6	365	0.069	< 0.5
D-718	[NAME1]	18	295	0.138	< 0.5
(.xls)	training	6	383	0.143	< 0.5
	crimecontrol	7	4	0.195	< 0.5
D-735	[NAME2]	3	22	0.006	< 0.5
(.msg)	[NAME3]	5	52	0.023	< 0.5
	NewYork	5	195	0.141	< 0.5
D-805	[NAME4]	5	4	0.038	< 0.5
(.txt)	Bermuda	5	1190	0.091	< 0.5
	[NAME5]	5	33	0.020	< 0.5

However, using only a single keyword consumes a lot of time for Investigators as this can delay the duration of the case. So it is better to use a combination of keywords and check the total time taken to process evidence items per file type. When tested for a combination of keywords FTK analysis time is worse than a single keyword search on the other side LES response time is in milliseconds it is more or less instant which makes investigation much faster and improves the quality of the research experience. Let us look at the results of both experiments.

Table 3. Total number of processed evidence items per file type

Document Type	Number of items per file type	
	FTK	*LES*
Email	1.585.500	1.641.063
Word documents	44.105	44.837
Spreadsheets	68.101	38.580
Presentations	6.548	2620

Table 2. Response time of combined keyword search

Document	*Keywords*	*FTK*		*LES*	
		Response Time(s)	*Retrieve Time(s)*	*Response Time(s)*	*Retrieve Time(s)*
D-195 (.msg)	discontinue, fraud, revenue	28	7.1	0.006	< 0.5
D-550 (.ppt)	scrubbing, blacklists, violation	24	5.2	0.138	< 0.5
D-718 (.xls)	[NAME1], training crimecontrol	10	18	0.195	< 0.5
D-735 (.msg)	[NAME2], [NAME3], NewYork	11	5	0.232	< 0.5
D-805 (.txt)	[NAME4], Bermuda, [NAME5]	16	4	0.004	< 0.5

In this way, this tool is a valuable asset for digital forensics.

Suggested Area for Further Research

We spoke about the abilities of the LES tool using NLP techniques and how it can manage those huge data of textual information in terms of speed the intended solution is quite significantly fast as well but there are a few areas where further research should be done, now its talk about those areas. Fraud patterns in white-collar crimes should have been researched further. When it comes to Fraud patterns need a past dataset where we can train the machine based on historical frauds and deploy it in a real environment where we can test the speed and accuracy of it and check if there are any chances of false positivity or adaptivity for the real world. Research should focus more on developing NLP modules where

that not only analyze the relationship between entities but also detect fraud patterns for example Recurrent Neural Networks can be used to train to detect fraud patterns and not only this, Graph neural networks can be used to detect suitable patterns across the networks. Researchers can further research and develop an algorithm that detects unusual clusters in a financial transaction potentially signaling online fraud.